**EXPERIMENTS**

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**Course Code:** ECA0901

**Course Name:** DIGITAL SIGNAL PROCESSING

# 1. BUTTERWORTH IIR FILTERS

## 1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

**CODE:**

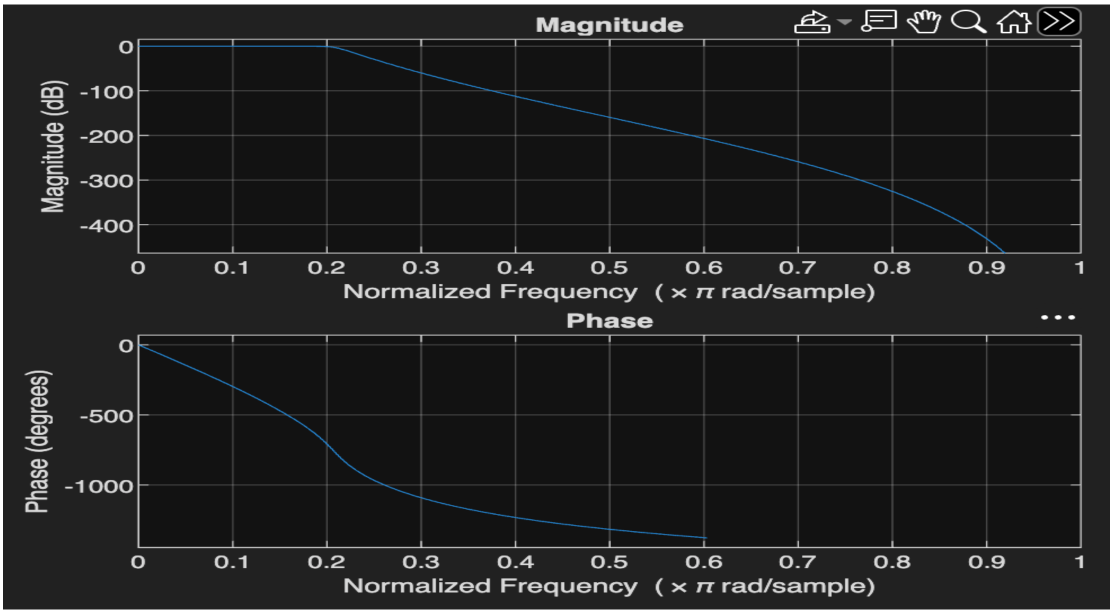
```
clc;
clear;
close all;
Fs = 10000;
Fp = 1000;
Fst = 1500;
[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a] = butter(N,Wn);
% Frequency Response
figure;
freqz(b,a);
% Impulse Response
[h,n] = impz(b,a,50);
figure;
stem(n,h,'filled');
grid on;
xlabel('Samples');
ylabel('Amplitude');
title('Impulse Response');
% Step Response
figure;
stepz(b,a);
% Pole-Zero Plot
figure;
zplane(b,a);
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

**OUTPUTS:**

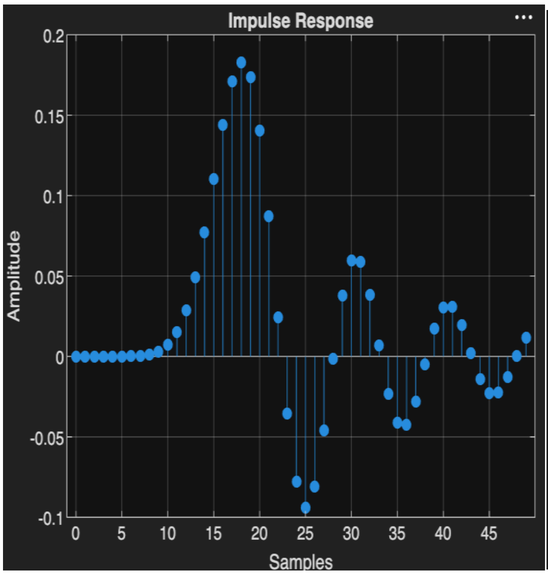
Objective 1:

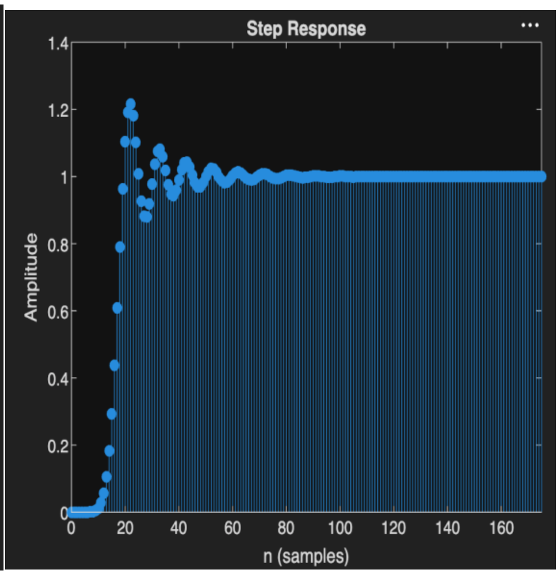
```
Minimum Filter Order (N) = 17
Cutoff Frequency (Wn) = 0.2083
```

Objective 2:

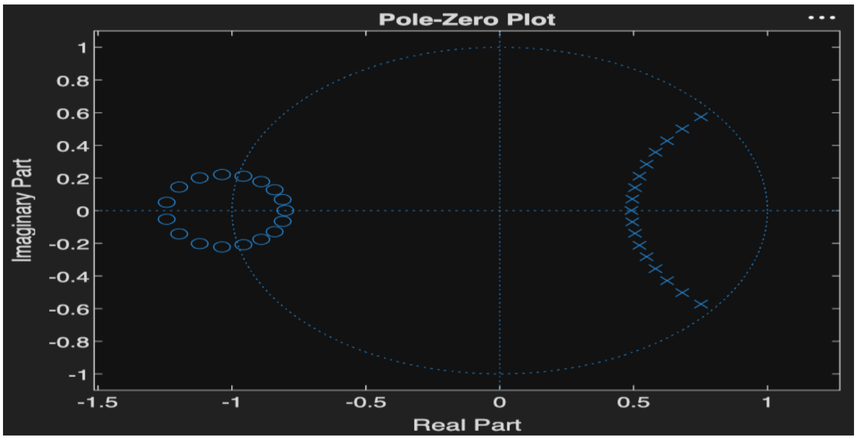



Objective 3:




Objective 4:




1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

**CODE:**

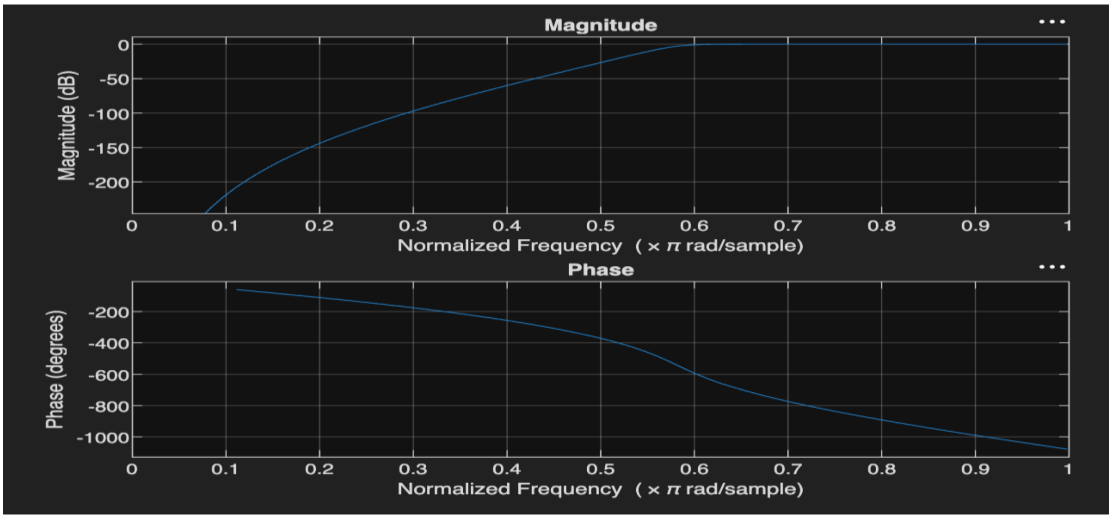
```
clc;
clear;
Fs=10000; Fp=3000; Fst=2000;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn,'high');
% Magnitude & Phase
freqz(b,a)
% Impulse Response
figure
[h,n]=impz(b,a,50);
stem(n,h), grid on
title('Impulse Response')
% Step Response
figure
stepz(b,a)
title('Step Response')
% Group Delay
figure
grpdelay(b,a)
title('Group Delay')
% Pole-Zero Plot
figure
zplane(b,a)
title('Pole-Zero Plot')
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

**OUTPUTS:**

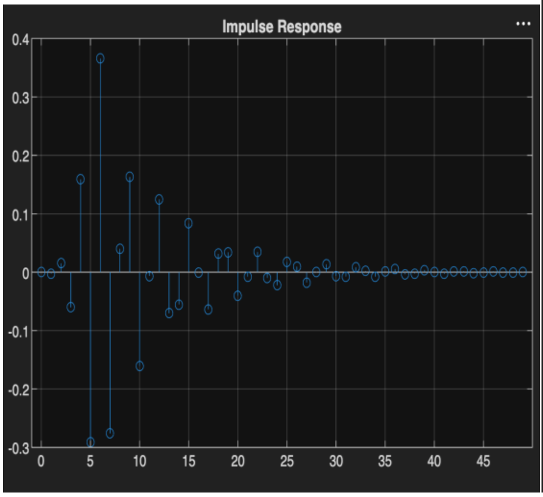
Objective 1:

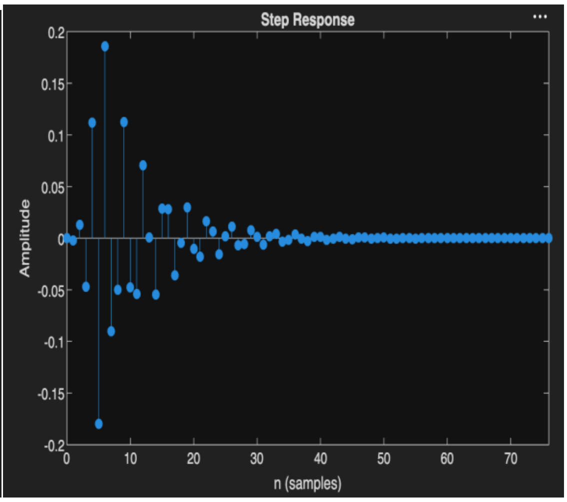
```
Minimum Filter Order (N) = 12
Cutoff Frequency (Wn) = 0.5807
```

Objective 2:

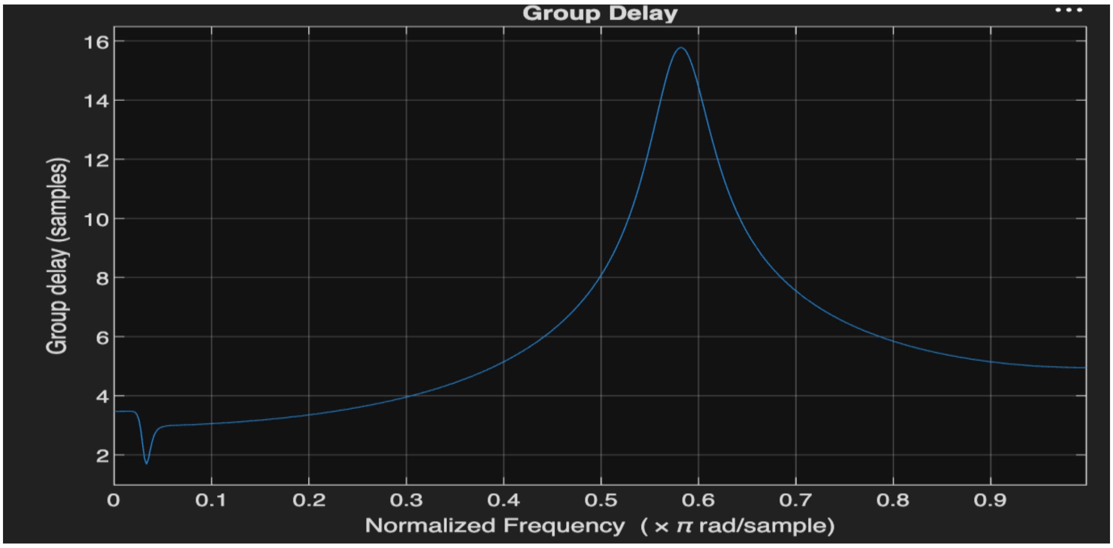



Objective 3:

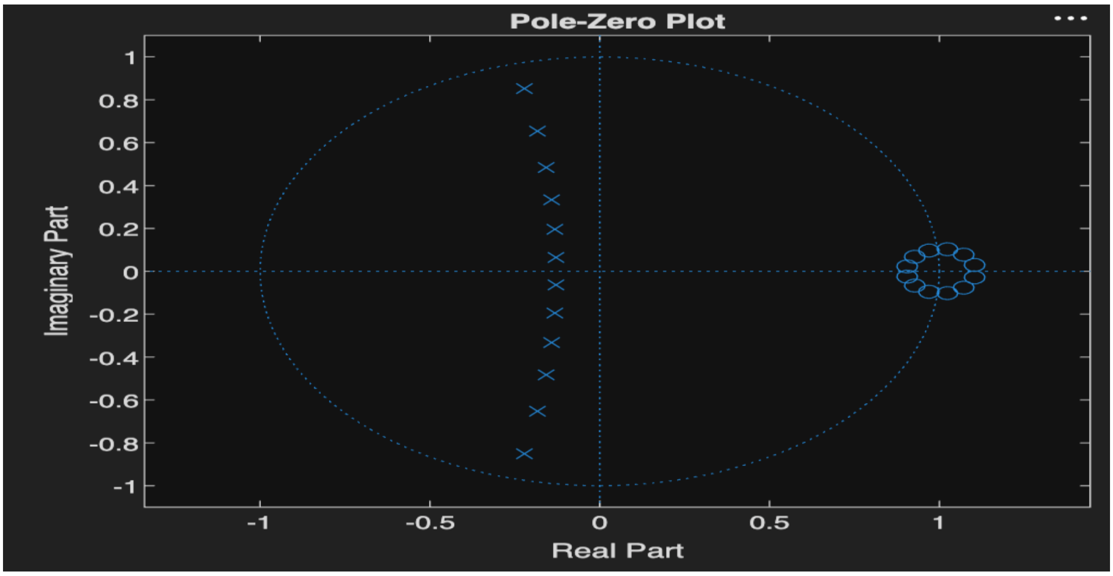



Objective 4:




Objective 5:



Pole-Zero Plot
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1